

A Surface-Content Decomposition of AI-Generated Text Detection Benchmarks: Evidence from Five Models on DAIGT V2 and HC3

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Abstract—A detector that scores 99% on a machine-generated-text benchmark may have modelled how machine language differs from human language, or how one corpus was punctuated and encoded, and the headline number does not distinguish them. We propose a decomposition that measures the difference. A surface-only model reads 47 orthographic features and never sees a word, a content-only model sees text after punctuation, casing and non-ASCII characters are stripped, and a full model is a fine-tuned transformer on raw text. Applied to DAIGT V2 and HC3 across five model families and eight configurations, the two corpora separate. On HC3 the surface-only and content-only arms are indistinguishable, 3.20% against 3.26% test error, so orthography alone matches content alone. On DAIGT V2 content is stronger by a factor of 7.9 in error, and a bag-of-words model on surface-normalised text comes within 0.0016 F1 of the best transformer, inside the measured seed range. Three controls change what the results mean. BERT emits identical token identifiers for strings differing only by the whitespace cue that dominates discussion of HC3, so it cannot represent that cue yet still reaches 0.9916 F1, making the cue sufficient in isolation but unnecessary in practice. A pipeline control establishes that the corresponding null is not a cleaning step that failed to run. A label-free control shows these models label random character strings as human in 100% of cases at 0.98 confidence, so an adversarial collapse to 0.3644 F1 cannot be read as a targeted vulnerability. Surface form is informative on both corpora, and we claim only that it reaches parity with content on one.

Index Terms—AI-generated text detection, benchmark evaluation, surface features, shortcut learning, tokenisation, ROC analysis

I. INTRODUCTION

Detectors of machine-generated text report accuracy at or above 97% on the benchmarks the field uses most, and frequently above 99% [1], [2], [3], [4]. Read directly, those numbers say the task is close to solved on the domains these corpora represent. Read carefully, they say something weaker, because a benchmark score measures separability of a particular corpus and not the capability a reader infers from it.

The difference matters because separability has more than one source. A detector can reach a high score by modelling how machine-generated language differs from human language, or by modelling how one corpus was assembled, punctuated and encoded. Both produce the same headline figure. Individual instances of the second have been documented, including a whitespace convention in HC3 [5], a length confound in the same corpus [6], punctuation inconsistencies

requiring standardisation [7], and prompt-specific collection shortcuts [8]. What is missing is a way to ask how much, in total, a given benchmark’s separability owes to surface form rather than content, and to compare benchmarks on that axis.

This paper proposes that measurement and applies it. We compare three arms on identical splits. A *surface-only* model reads 47 orthographic features and never sees a word. A *content-only* model sees text after punctuation, casing and non-ASCII characters have been stripped away. A *full* model is a fine-tuned transformer on raw text. The gap between the first two bounds how much of the benchmark a detector could pass without reading language at all.

Applied to DAIGT V2 and HC3 across five model families, the answer differs sharply between the two. On HC3 the surface-only and content-only arms are indistinguishable, 3.20% against 3.26% test error. On DAIGT V2 content is stronger by a factor of 7.9 in error, 0.99% against 7.86%. The same contrast appears in the model comparison, where the best classical configuration on DAIGT V2 falls 0.0007 F1 short of the best transformer, inside our measured seed range, but falls short by an error-rate factor of 16 on HC3.

We also report two negative results, because both correct claims that this analysis originally made and that the literature invites. First, the whitespace cue that dominates discussion of HC3 is sufficient in isolation yet unnecessary in practice. BERT’s tokeniser provably cannot represent it, emitting identical identifiers for strings that differ only in that respect, and BERT still reaches 0.9916 F1. Second, a character-level perturbation that drives DeBERTa from 0.9972 to 0.3644 F1 looks like a targeted vulnerability and is not one. The same model labels uniform random character strings as human in 100% of cases at 0.98 confidence, so the collapse is not distinguishable from a degenerate response to unreadable input without a label-free control. We report that requirement rather than the vulnerability.

Our contributions are the decomposition itself, as a reusable measurement over any human-versus-machine corpus. A second contribution is its application to two widely used benchmarks, with a complete eight-configuration model comparison and per-model ROC, area and confusion analysis. A third is the set of three controls, on tokenisation, on pipeline execution and on label-free inputs, each of which overturns a conclusion that the uncontrolled version of this study had reached.

II. RELATED WORK

Detection on these benchmarks. Guo et al. [1] introduce HC3 and report a RoBERTa detector at 99.82 F1 on English full text, establishing both the corpus and the accuracy regime the field has worked in since. Su et al. [2] show that regime is narrower than it appears, since moving from question answering to semantic-invariant tasks drops accuracy to roughly 91.7%, though their evaluation stays within ChatGPT-generated text. Zhou [3] reaches competitive accuracy on DAIGT V2 with classical models over character n-grams and no transformer at all, which is consistent with our DAIGT V2 finding but is reported without a surface-content separation that would explain it. Kubrusly et al. [9] likewise report strong bag-of-words results on HC3, and likewise do not isolate what the features encode. Ansary [4] and Lamsiyah et al. [10] report near-ceiling results on DAIGT-family data, in both cases with in-distribution evaluation only. Annepaka et al. [11] combine explicit linguistic features with a transformer and reach 99.45% on HC3, a design that mixes the two information sources our decomposition separates.

Documented artefacts. Tian et al. [5] identify the HC3 whitespace convention, observe that a detector could score well by exploiting it, and release a cleaning kit. Their contribution is the artefact and the remedy. They do not quantify how much of the corpus’s total separability surface form carries, and they do not test whether a given detector can represent the cue at all, which is the question our tokenisation control answers and which turns out to change the conclusion for BERT. Baidya et al. [6] document a length confound in HC3 and control it by length-matching, again treating one artefact in isolation. Ardeshtirfar [7] standardises punctuation and contractions before comparing handcrafted features to deep models, establishing the practice as methodologically necessary without measuring what it removes. Park et al. [8] show HC3-trained detectors rely on prompt-specific collection shortcuts and construct instructions exploiting them. Our decomposition is complementary to all four. Where they identify individual cues, we measure the aggregate, which is the quantity a practitioner comparing two benchmarks actually needs.

Robustness and its controls. Antoun et al. [12] attack HC3-trained detectors with misspellings and homoglyphs and report a fall from 99.88 F1 to 33.57% accuracy on a hand-built adversarial set. Huang et al. [13] report comparable degradation under systematic character perturbation. Neither reports a label-free control, and our results suggest such collapses cannot be separated from a generic degenerate response to unreadable input without one, since our own models label random character strings as human with near-maximal confidence. This is a methodological gap rather than a dispute about their measurements. Lai et al. [14] show that ensembling fine-tuned detectors preserves in-distribution accuracy while still degrading under distribution shift, so the fragility is not an artefact of single-model variance. Zhang et al. [15] step outside per-document classification altogether, framing detection as a

two-sample test, which sidesteps the single-document shortcuts studied here at the cost of needing a population. Borile and Abrate [16] take the complementary mechanistic route, ablating roughly twenty feed-forward neurons to improve out-of-distribution accuracy by up to 6.9 points, evidence that some in-distribution performance rests on localised corpus-specific features.

Cross-corpus evaluation. Alikhanov et al. [17] merge HC3 and DAIGT V2 into a single 124,195-sample corpus and evaluate under a topic-held-out split, reporting 82.87% to 88.86% accuracy against the near-ceiling numbers usual in-distribution. Their design asks whether one model can serve both corpora. Ours asks what each corpus is separable by, which is why the two results sit alongside rather than replacing one another. Lu et al. [18] show detectors can be evaded by prompting alone, a threat model orthogonal to the corpus-property question studied here.

III. METHOD

A. Datasets

DAIGT V2 [19] contains 44,868 argumentative student essays, 27,371 human-written from the PERSUADE corpus and 17,497 machine-generated by a mixture of 2023-era systems including GPT-3.5, Llama-2, Mistral-7B and Falcon-180B, and was assembled for a public detection competition [20]. The human side derives from the PERSUADE corpus of student argumentative writing [21]. HC3 [1] contains question-answer pairs from five English sources, contrasting human answers with responses from a single generator, GPT-3.5-Turbo-0301, over 85,449 rows [22]. Both were class-balanced for training, giving 34,994 DAIGT V2 rows and 53,806 HC3 rows, and partitioned once into fixed 80/10/10 splits used unchanged by every model reported here. Training partitions are balanced to within 0.002 of parity, which matters for interpreting the degenerate-prediction control in Section III-D.

A duplicate-detection pass over the unbalanced corpora finds DAIGT V2 essentially free of internal duplication, six rows in six groups, 0.01%, with no text appearing under both labels. HC3 carries 6,118 duplicate rows in 5,986 groups, 7.16%, plus 476 rows labelled as machine-generated that are collection artefacts rather than model output, typically stored API failure messages. At the smaller scale where every leaked row can be inspected by hand, eight of 1,200 HC3 test rows, 0.67%, also appear in training, and rescoring on the leakage-excluded subset moves F1 by at most 0.0010 without changing any ranking. We therefore report the contamination as a data-quality property rather than as an explanation of any accuracy figure.

B. Models and the combinations evaluated

Five model families are evaluated. Three are classical, Naive Bayes, logistic regression and a linear support vector machine, and each is run under two representations, bag-of-words and TF-IDF, giving six classical configurations per dataset. Earlier work on this corpus reported only the single winning representation per classical model at full scale. We report the complete

six-cell grid so the representation choice is visible rather than assumed. Two are transformers, `bert-base-uncased` and `microsoft/deberta-v3-base`, fine-tuned over a grid of learning rate $\{2e-5, 3e-5\}$, batch size $\{16, 32\}$ and weight decay $\{0.01, 0.1\}$, sixteen configurations per dataset, with the operating point selected on validation F1.

Classical models receive text through the normalisation pipeline described below. Transformers receive raw text truncated to 128 tokens, a constraint inherited from the study this corpus was built for, whose consequences we quantify in Section V rather than leave implicit.

C. The surface-content decomposition

The central instrument of this paper separates how much of a benchmark’s human-versus-machine separability is available from surface form alone and how much from content alone. Three arms are compared on identical splits.

The *surface-only* arm uses 47 orthographic features and never reads word identity. These cover punctuation character rates, whitespace behaviour including the count of spaces preceding punctuation, casing statistics, document and sentence length, non-ASCII and emoji rates, and digit rates. Because no lexical item enters the feature vector, this arm cannot access content by construction.

The *content-only* arm applies the surface normalisation used by the classical pipeline, lowercasing and replacing every character outside `[a-z\s]` with a space, then removing stop-words and lemmatising, before bag-of-words vectorisation. Punctuation, casing and non-ASCII characters cannot survive this transformation, so the arm cannot access the surface cues the first arm isolates.

The *full* arm is the fine-tuned transformer on raw text. Surface-only and content-only share one classifier family, logistic regression, so they are directly comparable to each other. The transformer differs in family and is reported as a reference point rather than as a matched arm, and we draw no conclusion from a transformer-to-classical difference alone.

D. Statistics and confound checks

Every reported figure comes from a committed script writing JSON, and every transformer baseline is read from the deployed checkpoint’s own record rather than from the maximum over its hyperparameter grid. These are not stylistic preferences. An earlier version of this analysis compared a grid maximum against a single fixed configuration and reported differences that did not exist, which is why baselines are now checkpoint-matched throughout.

We report error rate alongside F1 because at this accuracy ceiling F1 differences compress, and macro alongside weighted F1 so that no result can be produced by class skew. Seed spread over three seeds is reported as a range, and we treat it as an order-of-magnitude indication rather than a test statistic, since fine-tuning outcomes are known to vary materially with initialisation and data order [23], [24]. We do not run parametric tests at $n \leq 30$.

TABLE I
ALL MODEL CONFIGURATIONS, BOTH DATASETS, DEPLOYED CHECKPOINTS ONLY. ERROR RATE IS REPORTED BESIDE F1 BECAUSE F1 DIFFERENCES COMPRESS AT THIS CEILING. MACRO F1 EQUALS WEIGHTED F1 TO FOUR DECIMALS IN EVERY ROW, SO NO RESULT IS PRODUCED BY CLASS SKEW.

Model	Rep.	DAIGT V2		HC3	
		F1	err %	F1	err %
Naive Bayes	BoW	0.9591	4.09	0.8713	12.87
Naive Bayes	TF-IDF	0.9577	4.23	0.8670	13.28
Logistic Reg.	BoW	0.9893	1.07	0.9551	4.49
Logistic Reg.	TF-IDF	0.9857	1.43	0.9365	6.35
SVM	BoW	0.9870	1.30	0.9475	5.25
SVM	TF-IDF	0.9910	0.90	0.9449	5.51
BERT	raw	0.9916	0.84	0.9916	0.84
DeBERTa	raw	0.9917	0.83	0.9972	0.28

Three controls guard the specific claims made below. First, a *pipeline-fires* control. Our whitespace cleaning produces bit-identical predictions for BERT on HC3, which alone would be indistinguishable from cleaning that silently failed to run. The same pipeline applied to DAIGT V2 does change BERT’s predictions, since emoji removal is visible to its tokeniser, establishing that the pipeline fires and that the null is specific to whitespace. Second, a *tokenisation* control, comparing token identifiers for minimally different strings to establish what each tokeniser can represent. Third, a *label-free* control, scoring inputs that carry no human-versus-machine information at all, including uniform random character strings, token-shuffled text, punctuation-only strings and the empty string. Without this last control a confident prediction on corrupted input is indistinguishable from a learned response to it.

IV. RESULTS

A. A classical model matches a transformer on DAIGT V2 but not on HC3

Table I reports every model configuration on both datasets, and Fig. 1 gives the corresponding ROC curves and areas. The two benchmarks behave differently in a way no single headline number reveals. On DAIGT V2 the best classical configuration, an SVM over TF-IDF at 0.9910 F1, sits 0.0007 below the best transformer at 0.9917, a difference well inside the 0.0036 seed range we measure for this family of runs. On HC3 the same comparison is 0.9551 against 0.9972, an error-rate ratio of 16 to one. Whatever the transformers contribute on HC3, they contribute almost none of it on DAIGT V2.

Ranking by area does not reproduce ranking by F1. On DAIGT V2, BERT attains the higher area, 0.9996 against 0.9991, while DeBERTa attains the higher F1. We report both rather than the more favourable one.

B. On HC3, orthography alone matches content alone

Table II and Fig. 3 give the decomposition. On HC3 the two arms are indistinguishable, 3.20% error from orthography against 3.26% from content. Forty-seven features that never read a word do as well as a bag-of-words model over the whole vocabulary. On DAIGT V2 the same two arms separate

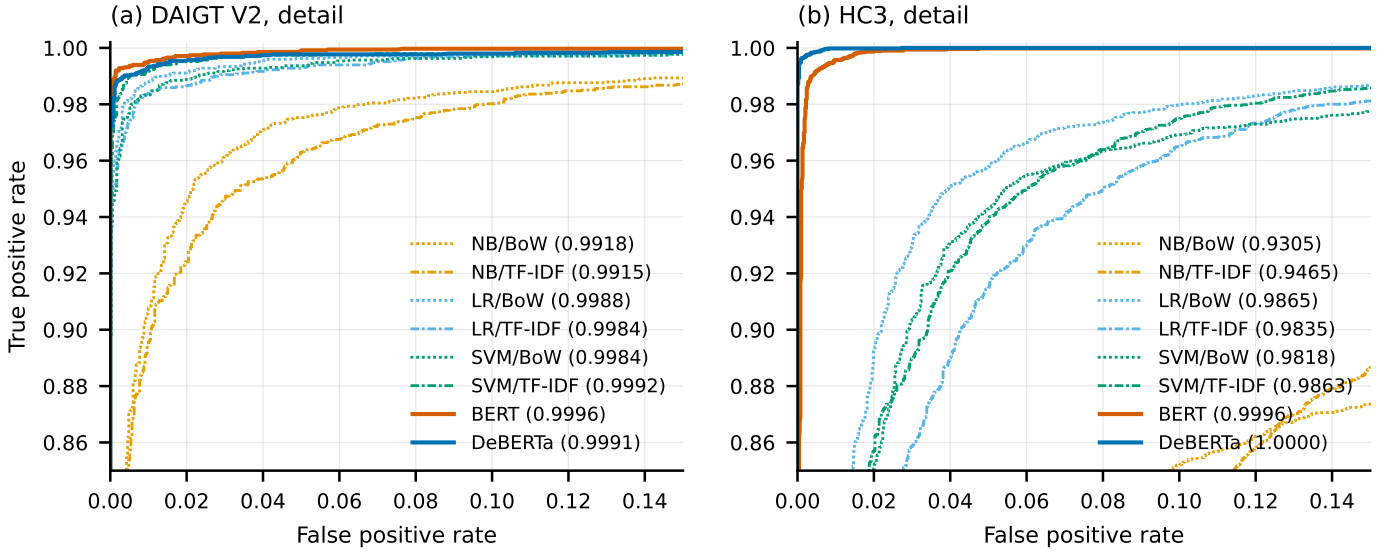


Fig. 1. ROC curves for all eight model configurations, restricted to the low-false-positive region where the models actually differ. Areas under the full curve are given in the legend. On DAIGT V2 (a) every configuration converges toward the corner and the classical models are close behind the transformers. On HC3 (b) the two transformers separate sharply from all six classical configurations. Colours are colourblind-safe and each series carries a distinct dash pattern, so the ordering survives greyscale reproduction.

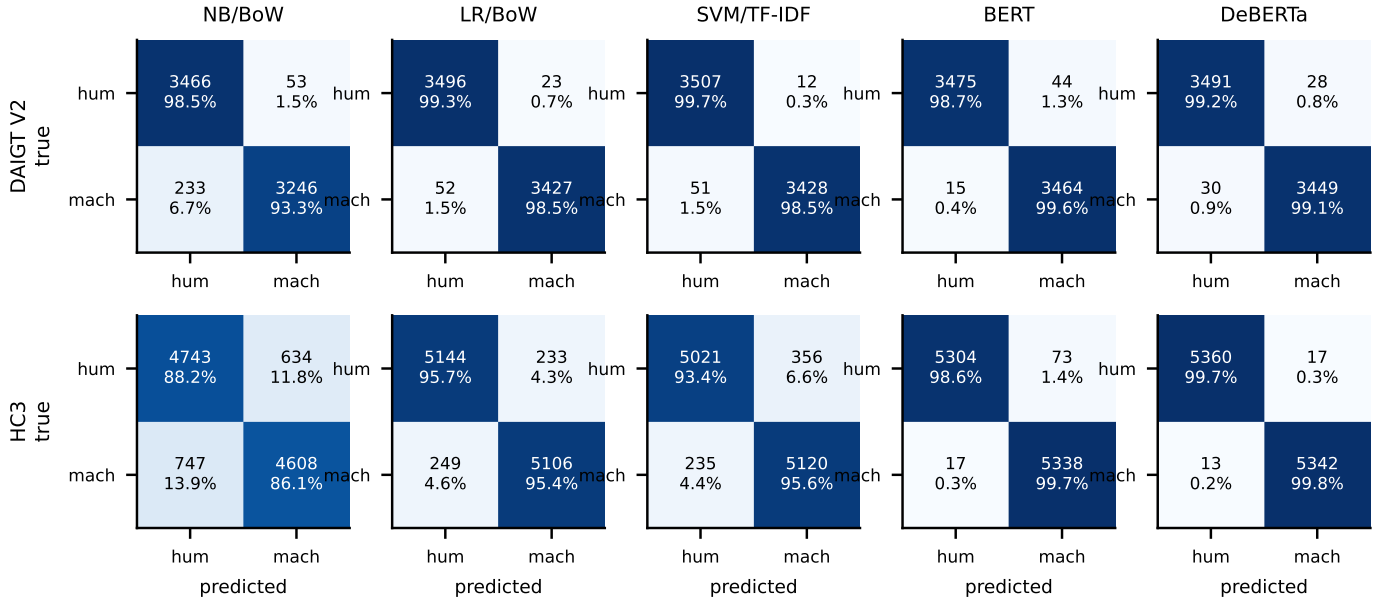


Fig. 2. Confusion matrices for the best configuration of each model family, both datasets, as counts and row-normalised percentages. Naive Bayes on HC3 is the only cell with substantial error in both directions, and every other cell concentrates its error on one side.

TABLE II
SURFACE-CONTENT DECOMPOSITION. SURFACE-ONLY USES 47 ORTHOGRAPHIC FEATURES AND NEVER READS A WORD. CONTENT-ONLY STRIPS PUNCTUATION, CASING AND NON-ASCII BEFORE VECTORISING. THE TRANSFORMER IS A REFERENCE POINT, NOT A MATCHED ARM.

Arm	DAIGT V2		HC3	
	F1	err %	F1	err %
surface-only	0.9214	7.86	0.9680	3.20
content-only	0.9901	0.99	0.9674	3.26
full transformer	0.9917	0.83	0.9972	0.28

by a factor of 7.9 in error, 7.86% against 0.99%, with content the stronger by a wide margin.

Two qualifications belong with this result rather than after it. First, surface form is informative on both benchmarks. DAIGT V2’s surface-only arm reaches 0.9214 F1, far above chance, so the finding is not that one dataset is clean and the other is not. It is that only on HC3 does surface form rise to parity with content. Second, an earlier version of this analysis used a single hand-chosen threshold on one whitespace feature and concluded that DAIGT V2 was separable at chance. That conclusion was an artefact of the instrument, not a property of

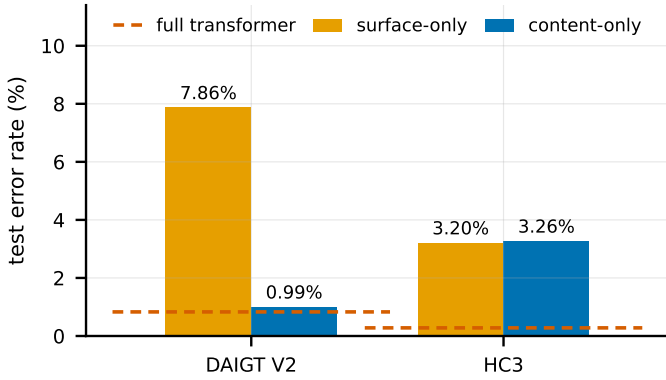


Fig. 3. Test error for the surface-only and content-only arms, with the best full transformer marked. On HC3 the two arms coincide. On DAIGT V2 content is stronger by a factor of 7.9 in error.

the dataset, and a fitted multi-feature model overturns it. We report the fitted result and retract the threshold one.

The content-only arm also bounds what the transformers add. On DAIGT V2 content-only reaches 0.9901 against the transformer’s 0.9917, a difference inside the seed range, so the transformer adds nothing measurable over a bag-of-words model on surface-normalised text. On HC3 the transformer improves on either single arm by roughly 0.03 F1, consistent with combining information that neither arm holds alone.

C. A model that cannot represent the cue reaches 0.9916 regardless

The most-discussed HC3 artefact is a space preceding punctuation in human answers [5], present in our copy at 10.745 occurrences per human document against 0.013 per machine document. It is natural to assume detectors exploit it. Tokenisation makes that assumption testable rather than rhetorical.

BERT’s WordPiece tokeniser splits on punctuation irrespective of adjacent whitespace. For the pair "the answer is simple ." and "the answer is simple." it emits identical identifier sequences, and does so for three of three such pairs we tested. DeBERTa’s SentencePiece tokeniser encodes leading whitespace explicitly and distinguishes all three. BERT therefore cannot represent this cue at all, and still reaches 0.9916 F1 on HC3. The cue is sufficient in isolation and unnecessary in practice.

This is also why removing it changes nothing for BERT. Applying whitespace cleaning to HC3 yields predictions bit-identical to the uncleaned run, which taken alone would be indistinguishable from a cleaning step that never executed. The same pipeline applied to DAIGT V2, where it removes emoji that WordPiece does represent, does change BERT’s predictions. The pipeline fires, and the null is specific to whitespace.

We do not attribute the BERT-to-DeBERTa difference on HC3 to this cue. The two models differ in tokeniser, architecture, pretraining corpus and parameter count, and this experiment separates none of those.

D. The adversarial collapse is not distinguishable from a degenerate response

Perturbing HC3 test text with character-level typos at 10% of alphabetic characters drives DeBERTa from 0.9972 to 0.3644 weighted F1. The confusion matrix is one-directional, $[[5376, 1], [5207, 148]]$, with human documents still classified perfectly and machine documents reassigned to human. Read alone this looks like a targeted vulnerability. It is not what the control shows.

On inputs carrying no human-versus-machine information, the same model assigns *human* to uniform random character strings in 100% of cases at mean confidence 0.98, and to token-shuffled and punctuation-only text in 100% of cases. All four models label the latter two as human between 94.8% and 100% of the time. The behaviour is specific to neither HC3 nor the attack.

Three candidate mechanisms were tested and none survives. Majority-prior collapse is excluded because the training partitions are balanced to within 0.002. Cue injection is excluded because the perturbation does not create the whitespace artefact, moving machine-document counts from 0.000 to 0.005 per document while homoglyph substitution moves them not at all. A learned association between disfluency and human authorship is unsupported, because subword fertility runs the wrong way, 1.1192 for HC3 human text against 1.1980 for machine text.

We therefore decline to read our own perturbation numbers as robustness measurements, and note that the same caution applies to any comparable collapse reported without a label-free control. What the experiment establishes is a requirement on future work, not a property of these benchmarks.

E. Cross-dataset transfer degrades in both directions

Training on one benchmark and testing on the other, over three seeds, costs between 0.0833 and 0.2019 mean F1 depending on direction and model, worst for DAIGT-V2-trained BERT at 0.9921 falling to 0.7902. The asymmetry reverses between model families, so with three seeds per cell we report the magnitudes and draw no conclusion from the direction.

V. DISCUSSION AND CONCLUSION

The decomposition separates two benchmarks that a headline accuracy figure does not. On HC3, orthography alone matches content alone, 3.20% against 3.26% test error, and a fine-tuned DeBERTa improves on either by roughly 0.03 F1. On DAIGT V2, content is stronger than orthography by a factor of 7.9 in error, and a bag-of-words model over surface-normalised text comes within 0.0016 F1 of the best transformer, inside our measured seed range. A practitioner choosing an evaluation corpus, or a reviewer reading a reported score, cannot recover that distinction from the score itself.

What the measurement does not license is a claim that either corpus is clean. DAIGT V2’s surface-only arm reaches 0.9214 F1, far above chance, so surface form is substantially informative on both. The finding is comparative, that surface reaches parity with content on one corpus and not the other,

and it is bounded by the arms we built. A richer surface featurisation would raise both surface numbers, and a stronger content model would raise both content numbers, so the arms bound rather than partition the separability.

Two results here began as the opposite conclusion, and both are worth stating for what they say about method. The whitespace cue in HC3 is the most-discussed artefact in this literature, and it is sufficient in isolation. It is also unnecessary, since BERT’s tokeniser cannot represent it and BERT reaches 0.9916 F1 regardless. A cleaning experiment that reports no change for BERT is therefore uninterpretable without the accompanying evidence that the pipeline runs at all, which we supply by showing the same pipeline does change BERT’s DAIGT V2 predictions where the artefact removed is one WordPiece encodes. Similarly, a character perturbation drives DeBERTa from 0.9972 to 0.3644 F1 with a one-directional confusion matrix, which reads as a targeted vulnerability until the same model is shown to label uniform random strings as human in 100% of cases at 0.98 confidence. We report the requirement for a label-free control rather than the vulnerability, and we suggest that comparable published collapses be read with that control in mind.

Several limits bound every number above. The transformers see 128 tokens, which covers a median HC3 document but only about 31% of a median DAIGT V2 essay, so the DAIGT V2 transformer results describe classification from an essay’s opening and the cross-corpus comparison is not like-for-like in the information each model receives. Seed spread is estimated from three seeds and reported as a range, not a test statistic, and one such estimate moved by 31% when its runs were repeated, which is why no claim here rests on a variance ratio. HC3 contains output from one generator collected at one time, both corpora are English, and the classical arms share one preprocessing pipeline whose stopword removal and lemmatisation remove more than punctuation, so the content arm is a lower bound on content-carried separability rather than a clean isolate. We evaluate two corpora and five model families, and nothing here establishes that the pattern holds beyond them.

The measurement itself is the portable part. It requires no new annotation, runs in minutes on a laptop, and yields one number per corpus that a headline accuracy cannot express. Reporting it alongside detection results would let a reader see, before trusting a 99% figure, how much of the corpus a model could pass without reading the language.

DATA AND CODE AVAILABILITY

Both datasets, DAIGT V2 and HC3, are public, and code is released for reproduction. The link is withheld for blind review.

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